

CubeSat Prototype: A Low-Cost 1U Platform for Deforestation Monitoring

Arnav Venkatesh
Grade 12, NHVPS, Bangalore
May 2026

Abstract — This report documents a self-built, bench-tested 1U CubeSat prototype combining a live camera feed, attitude sensing, magnetorquer-based orientation control, and a wireless dashboard, all running on a single ESP32 module. The payload was integrated with a computer vision model to demonstrate deforestation detection. The project was built solo over one year for under \$71 in electronics and printing costs, and is documented openly on GitHub for other student builders.

1 Motivation and Origin

This project began after watching a video by Mark Rober on building CubeSats. I did not understand how a satellite could reorient itself in orbit without chemical thrusters, which led me to research magnetorquer-based attitude control. That research resulted in a published technical review proposing a hybrid bang-bang and PD control approach. I wanted to test elements of that idea in a physical prototype, which became this project.

Building it stretched what I knew. Getting a live camera feed, telemetry, and motor control to run reliably together on one microcontroller failed many times before it worked. Through that process I learned to debug hardware and firmware, accept trade-offs, and design around what could realistically be built and tested at home.

2 Objectives

- Demonstrate a functional 1U CubeSat payload: live camera, attitude telemetry, and magnetorquer control on a single ESP32
- Validate the imaging pipeline against a deforestation classification model
- Design and iterate a 1U mechanical frame through CAD
- Document the build, including its limitations, transparently

3 System Overview

The prototype is built around an ESP32-CAM as the central controller, handling imaging, sensing, and actuation simultaneously.

Component	Role
ESP32-CAM (AI Thinker)	Main compute, camera capture, WiFi
MPU6050	3-axis accelerometer and gyroscope for pitch/roll
L298N Motor Driver	H-bridge control for magnetorquer coils
Hand-wound copper coils (×2)	Attitude actuation, X and Y axis
Battery pack + buck converter	7.5V step-down to regulated 5V

A key hardware constraint was that the ESP32-CAM's default I2C pins (IO12/IO13) conflict with the onboard SPI flash chip, causing boot failures. The IMU was moved to IO14/IO15 to resolve this. The firmware serves three endpoints: a live MJPEG stream, a JSON telemetry feed, and a combined web

dashboard, split across two HTTP ports to prevent browser request conflicts observed on Safari.

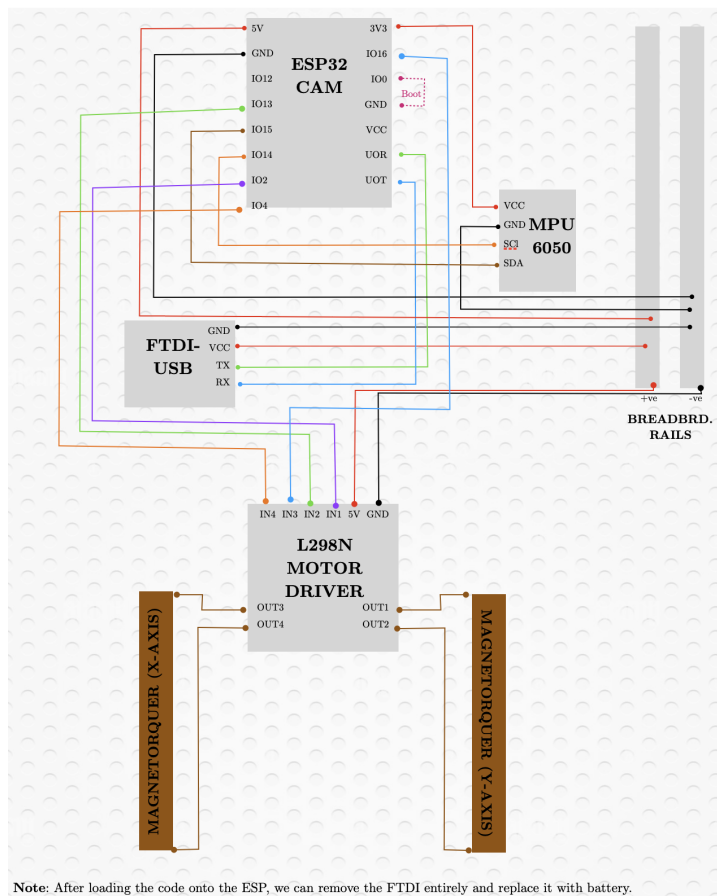


Figure 1: Full wiring schematic

4 Software and Web Dashboard

Rather than requiring a laptop connection to read sensor data, the firmware hosts a browser-based dashboard showing the live camera feed alongside real-time pitch, roll, and acceleration values, refreshed via JavaScript polling every 500 ms. This made the system fully wireless: once powered by battery, any device on the same WiFi network can monitor the satellite without a physical connection.

5 Payload Integration: Deforestation Detection

To demonstrate a practical use case, the live ESP32 camera feed was used as input to a separately trained computer vision classifier (documented in a companion repository) that flags deforested versus intact land cover from satellite-style imagery. In this bench demonstration, printed or on-screen satellite images were shown to the CubeSat's camera and classified in real time, showing how the payload could function if deployed with an actual orbital camera.

6 Engineering Challenges

- **Flash/I2C pin conflict:** Required migrating the IMU to non-default GPIOs after repeated boot failures
- **Upload reliability:** ESP32-CAM boot-mode uploading required a specific, repeatable IO0-to-GND and power-cycle sequence
- **Subsystem integration:** Combining camera, IMU, and motor driver firmware on one microcontroller required careful resource and pin management
- **Concurrent web requests:** Splitting the camera stream and telemetry data across two ports

resolved browser-side request failures

7 Results

Subsystem	Status
Camera stream	Working, live MJPEG over WiFi
Attitude sensing	Working, pitch/roll via accelerometer
Web dashboard	Working, no laptop required
Magnetorquer control logic	Working, bench tested
Battery-powered operation	Working, fully wireless
1U mechanical frame	Designed and 3D printed
CV payload demo	Working, bench-level integration

Total project cost was approximately 6,757 (\$71), with the electronics stack alone under \$21.

8 Limitations

This is a bench prototype, not a flight-ready satellite, and the following limitations are worth stating plainly. WiFi is used for telemetry as a pragmatic substitute for real radio hardware and only works on a local network. The frame has not been tested in vacuum, microgravity, or radiation conditions, largely because space-grade materials are outside this project's budget. The magnetorquer coils, hand-wound from copper wire, are too weak to physically reorient the frame, even though the control logic issues correct commands. Attitude control logic uses simple on/off switching rather than the hybrid algorithm proposed in the published review. The CV demo uses reference imagery shown to the camera rather than genuine orbital captures. No solar charging is implemented.

9 Future Work

- Implement the hybrid PD magnetorquer algorithm from the published review
- Replace WiFi telemetry with a LoRa or UHF radio module
- Conduct thermal and vacuum stress testing
- Design a fully enclosed frame with integrated panels

10 Conclusion

This project turned a question I could not answer, how a satellite moves without thrusters, into a working, if modest, physical system. It did not go as planned at every stage, and it is not going to space. But it worked, it taught me to debug real hardware failures, and it showed me that engineering is mostly about building the best version of something within real constraints, not the version you first imagined.

11 Build Gallery

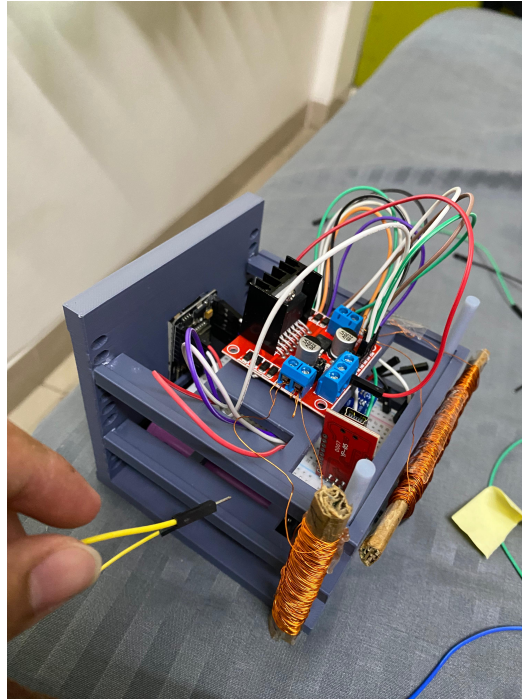


Figure 2: In-progress build

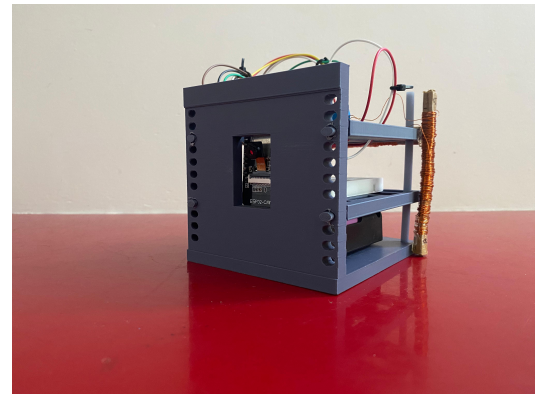
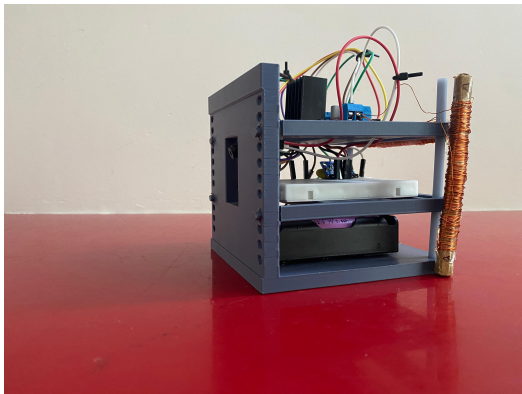


Figure 3 & 4: Completed prototype, full profile

References

1. Venkatesh, A. *CubeSat Prototype Repository*. <https://github.com/arnavvenkatesh/cubesat-prototype>
2. Venkatesh, A. *Satellite Deforestation Detection Repository*. <https://github.com/arnavvenkatesh/satellite-deforestation-detection>
3. Venkatesh, A. *Optimizing CubeSatellite Orientation Using Magnetorquers*. OneYoungIndia, 2025. [Review Paper Link](#)